

8) The Problem of Running Average & Its Expectation

- Define running average $m_t = \delta m_{t-1} + (1 - \delta)g_t$, $m_0 = 0$.
- Finite geometric series indicates: if data is stationary,

$$E[m_t] \approx E[g_t](1 - \delta^t) + \zeta',$$

indicating early **bias towards 0** (due to initialization at 0). The same reasoning applies to the second-moment running average.

The bias arises from initialization ($m_0 = 0$), causing early “averages” to be influenced by **zero padding**; the goal of correction is to eliminate this “start-up bias.”

9) Running-Average Bias Correction

$$\hat{E}[(\partial_w L)^2]_{t-1} = \frac{E[(\partial_w L)^2]_{t-1}}{1 - \gamma^{t-1}}, \quad \hat{E}[\partial_w L]_{t-1} = \frac{E[\partial_w L]_{t-1}}{1 - \delta^{t-1}}.$$

The correction effect is **largest** at small t .

These denominators serve as the **normalization constants** for the geometric weights (offsetting the bias from “starting at 0”), similar to the standard unbiased correction for exponential weighted moving averages (EWMA).

10) Adam: Adaptive Moment Estimation

$$\boxed{w_t = w_{t-1} - \alpha \frac{\hat{E}[\partial_w L]_{t-1}}{\hat{E}[(\partial_w L)^2]_{t-1} + \varepsilon}}, \quad \begin{aligned} \hat{E}[(\partial_w L)^2]_{t-1} &= \frac{E[(\partial_w L)^2]_{t-1}}{1 - \gamma^{t-1}}, \\ \hat{E}[\partial_w L]_{t-1} &= \frac{E[\partial_w L]_{t-1}}{1 - \delta^{t-1}}. \end{aligned}$$

Typical: $(\alpha = 10^{-3}, \gamma = 0.999, \delta = 0.9, \varepsilon = 10^{-7})$.

11) Learning Rate Scheduling

- Many popular methods can be viewed as GD with a shaped step:

$$w^t = w^{t-1} - \frac{\alpha}{\sqrt{\hat{E}[(\partial_w L)^2]_{t-1} + \varepsilon}} \hat{E}[\partial_w L]_{t-1}.$$

- Why schedule the learning rate ourselves?
 - Beginning:** use a **large** LR to increase speed.
 - End:** use a **small** LR to improve accuracy.

Learning rate decay schedulers (as on the slides):

- “Old days” manual: reduce LR by a factor of 10 every **XX** epochs.
- Factor:** $\alpha_{t+1} = \max(\alpha_{\min}, \alpha_t \cdot \tau)$
- Multi-factor (milestones):** $\alpha_{t+1} = \alpha \cdot \tau$ every several epochs / at milestones.
- Cosine:**

$$\alpha_t = \alpha_T + \frac{\alpha_0 - \alpha_T}{2} \left(1 + \cos \left(\frac{\pi t}{T} \right) \right)$$

- Warmup:** for unstable optimization, start with a **small** LR then ramp to target LR.

